

Ophthalmology

Departmental Objectives

The objective of this course is to provide need-based education so as to produce a quality doctor who will be able to

- deal with common ocular ailments
- identify, give initial management & refer ocular emergency cases appropriately
- provide leadership in the sphere of primary eye care in the country as well as abroad.

To achieve the above mentioned departmental objectives, the following learning objectives will be required:

List of Competencies to acquire:

1. Measure visual acuity of adult and children, a. unaided b. with pin hole c. with glass;
2. Examine color vision & examination of visual field (confrontation method)
3. Examine ocular movement and alignment; assessment of pupillary light reflex (direct and consensual)
4. Perform direct ophthalmoscopy.
5. Perform digital tonometry.
6. Perform Regurgitation test of lacrimal sac.
7. Perform Fluorescein dye test, irrigation of conjunctival sac & installation of eye drops/ointment.
8. Perform eversion of upper lid & removal of conjunctival foreign body.
9. Diagnose and give treatment of bacterial conjunctivitis, vitamin A deficiency disease (night blindness, Bittot's spot, xerophthalmia), initiate treatment of minor trauma, correction of simple presbyopia and referral of difficult cases.
10. Diagnose and initiate treatment and referral of ocular emergency cases:
a. trauma, b. painful red eye. c. corneal ulcer/keratitis, d. corneal foreign body, e. acute dacryocystitis.
11. Diagnose and referral for specialist management: cataract, chalazion, pterygium, leucocoria of children, squint, cases with reduced vision

Fundamentals and principles of ophthalmology

Goal: The students will have the overall understanding of external and internal ocular structures of the normal human eye and will be able to perform the eye examination in normal and disease conditions.

Topic Specific objectives:

At the end of the teaching of the course the students will be able to:

- describe normal ocular anatomy.
- obtain detail ocular history.
- measure and record visual acuity in adults and children.
- assess pupillary reflexes.
- evaluate ocular motility.
- use the direct ophthalmoscope for gross assessment of red reflex, the optic disc and fundus examination.
- perform and evaluate visual fields by confrontation.

Specific contents in this subject will include:**A. Ocular Anatomy.**

Students should be able to define gross anatomy of the eyeball& adnexa

1. Eyelids.
2. Extraocular muscles.
3. Lacrimal apparatus
4. Conjunctiva.
5. Cornea
6. Sclera.
7. Anterior chamber
8. Iris
9. Pupil.
10. Lens
11. Ciliary body
12. Posterior chamber
13. Vitreous cavity.
14. Retina
15. Optic disc.
16. Macula.
17. Choroid.
18. Optic nerve.

Learning Objectives**A. Knowledge components:**

Students will be able to describe:

1. basic ocular anatomy
2. concept of measuring visual acuity without correction ,with pinhole and with correction
3. the importance of assessing ocular motility in the six cardinal positions of gaze and ocular alignment in primary position
4. the basic function of ophthalmoscope
5. importance of dilatation of pupil for fundus examination
6. abnormal fundal appearance in diabetic and hypertensive retinopathy
7. the concept of measuring intraocular pressure
8. the technique of determining the peripheral visual field by confrontation method
9. referral guideline

B. Skill Components:

At the end of the course, the students will able to demonstrate the skill of:

1. examination of each eye individually.
2. test V/A each eye individually and with pinhole.
3. evaluation of the position of the lids, and inspection of the conjunctiva, sclera, cornea and iris with a penlight.
4. examination of the pupil and assessment of the pupillary reaction.
5. ocular motility test in six positions and cover test
6. manual sac regurgitation test
7. assessment of intraocular pressure by digital method
8. performing visual field assessment by confrontation method
9. eversion of the upper lid and examine for the presence of foreign bodies
10. fluorescein dye test and its interpretation.
11. performing direct ophthalmoscopy and identify structures eg. optic disc, macula, and major vessels.

C. Attitude component:

Students will show continuous interest in gaining information in the subject and at the end of the teaching; they will be able to demonstrate the following:

- a. A patient-centered role:
- b. Scientific Integrity:
- c. Ethical medical Professional Behavior:
- d. Dedication to Continuous Learning:

Learning will be facilitated by:

Active participation in the

- a. Classroom discussion
- b. Completion of assignments
- c. Formal presentations in tutorials.
- d. Self-initiated independent thinking, presentation skill.

Evaluation:

Students will be evaluated by

- a. Written examination(Short Essay test and MCQ test)
- b. Formal and informal observations by instructor
- c. Terms examinations
- d. Final assessment together with other topics in the final Professional MBBS examination.
- e. Class and ward attendance

Remediation during training:

- 1. The course coordinator will review the student's performance and will:
 - i. Identify any specific deficits
 - ii. Document all areas requiring remediation or additional concentration.
 - iii. Provide additional recommendations for remediation of specific lackings.

Method of teaching:

- a. Didactic lecture
- b. In-class group session
- c. Clinical class in the hospital out-patient, in-patient and Operation Theatre settings
- d. Problem based discussion.

Materials

Models, power point presentation will be provided and students will get copies of handout whenever available.

Learning Objectives and Course Contents in ophthalmology

Learning Objectives	Contents	Teaching Hours
Student will be able to: 1. describe the anatomy of orbit and its contents 2. describe gross anatomy of the extra ocular muscles 3. diagnose orbital cellulitis, proptosis, squint /deviation and asymmetry and refer to specialist care 4. list the conditions for further referral to specialist care	Orbit: 1. Gross Anatomy: a. Bones of the orbit constituting walls, roof and floor b. Contents of the orbit 2. Clinical examination of orbital disease: 3. Orbital diseases: a. Orbital cellulitis b. Proptosis	2 hrs
Students will be able to 1. describe gross anatomy of the lid 2. describe surgical steps of chalazion operation. 3. demonstrate the skill of step wise clinical examination, 4. describe diagnosis and treatment procedure of the followings; Sty, chalazion and blepharitis. 5. identify and refer the following: Trichiasis, ptosis, ectropion, entropion, chalazion 6. perform eversion of the lid.	Eye lids: 1. Gross Anatomy of the eye lid & its disease 2. Clinical Examination procedure a. Corneal light reflex & palpebral fissure height b. Visual inspection of eyelids and periocular area. 3. Diseases of Lid a. Malpositions.(definitions) i. Trichiasis ii. Ptosis iii. Ectropion iv. Entropion. b. Inflammations. i. Sty ii. Chalazion iii. Blepharitis iv. Internal hordeolum	2 hrs

Learning objectives	Contents	Teaching Hours
Students will be able to : 1. describe gross anatomy of conjunctiva 2. name diseases of the conjunctiva 3. describe surgical steps of pterygium operation. 4. examine the conjunctiva 5. diagnose and manage of viral, bacterial, allergic conjunctivitis & ophthalmia Neonatorum 6. diagnose pterygium and refer for surgical management 7. remove superficial conjunctival foreign body	Conjunctiva: 1. Gross Anatomy of the Conjunctiva & its diseases: 2. Examination procedure for conjunctiva 3.Disease of conjunctiva: a. Conjunctivitis - Bacterial - Viral - Allergic b. Ophthalmia neonatorum c. Trachoma (Gross idea) d. Pterygium 4.Precautionary measures:	2 hrs
Students will be able to: 1. describe the anatomy of lacrimal apparatus 2. describe production, and functions of tear. 3. describe steps of sac patency test with interpretation 4. describe symptoms, signs of lacrimal sac diseases. 5. diagnose and manage lacrimal sac diseases. 6. mention indication, contraindication and major complications of DCR and DCT 7. perform digital regurgitation test 8. perform digital massage in congenital nasolacrimal duct obstruction. 9. initiate treatment of acute & chronic dacryocystitis, and congenital nasolacrimal duct obstruction, and referred to an ophthalmologist	Lacrimal Apparatus: 1. Gross Anatomy of the Lacrimal Apparatus& its diseases: 2. Physiology: Function of tear. 3.Examination Technique: 4. Lacrimal sac disease: a. Actuate dacryocystitis. b. Lacrimal sac abscess c. Chronic dacryocystitis. d. Congenital nasolacrimal duct obstruction	2 hrs

Learning objectives	Contents	Teaching Hours
Students will be able to 1. describe gross anatomy of the fibrous coat of the eye 2. describe supply of nutrition to cornea and maintenance of its transparency 3. describe steps of performing fluorescein dye test. 4. describe Keratoplasty 5. examine cornea 6. perform fluorescein dye test (to detect corneal epithelial defect) 7. remove superficial nonimpacted corneal foreign body 8. diagnose, and initiating treatment of corneal ulcer, keratitis and appropriate referral	Cornea and sclera: 1. Gross anatomy of cornea and sclera 2. Physiology: a. Maintenance of nutrition& transparency of cornea b. Function of cornea c. Tear film 3. Diseases of cornea a. corneal ulcer b. keratitis c. Keratoplasty (Gross idea)	3 hrs
Student will be able to 1. describe the parts of uveal tract. 2. describe diseases of uveal tract, symptoms, signs and management of acute iritis & endophthalmitis 3. identify circumcorneal / ciliary congestion 4. perform pupil examination 5. identify ciliary tenderness 6. diagnose, initiation of treatment of iritis, endophthalmitis and appropriate referral.	Uveal tract 1. Gross Anatomy 2. Diseases of uveal tract a. Anterior uveitis/uveitis b. Endophthalmitis c. Panophthalmitis	2 hrs

Learning objectives	Contents	Teaching Hours
Students will be able to: 1. describe clinical features of age related cataract 2. describe stages of senile cataract 3. mention indications of cataract surgery 4. mention complications of untreated cataract 5. perform the preoperative evaluation 6. state ECCE, SICS and phaco surgery. 7. mention Complications of cataract operation 8. state Advantage of IOL implantation over spectacle 9. demonstrate the skill of diagnosis of cataract and referral to proper ophthalmologist	Lens and cataract: 1. Gross Anatomy: 2. Physiology: Accommodation 3. Disease of the lens a. Cataract b. Pseudophakia c. Aphakia 4. Management of cataract: a. Cataract surgery (Gross idea) b. Intraocular lens and its advantage (Gross idea) 5. Referral criteria of a cataract case	3 hrs
Student will be able to: 1. describe anatomy of the anterior chamber and anterior chamber angle 2. describe production circulation and outflow of the aqueous humor 3. define and classify glaucoma. 4. describe Symptoms, signs and management of POAG, PACG and congenital glaucoma 5. demonstrate the skill of: a. taking history of glaucoma patients. b. digital tonometry. c. conformation test d. direct ophthalmoscopy 6. diagnose and provide initial management of PACG and early referral. 7. counseling of all glaucoma patient regarding blinding nature of disease & necessity of life long regular treatment & follow up	Glaucoma: 1. Gross Anatomy 2. Physiology a) Production, circulation and outflow of the aqueous humor. b) Intra ocular pressure and factors influencing IOP. 3. Classification of glaucoma. 4. Disease.(gross aspect) a) Primary angle closure glaucoma i) Risk factors ii) Symptoms iii) Signs iv) Management b) Primary open angle glaucoma: i) Risk factors ii) Symptoms c) Congenital glaucoma i) Genetics ii) Symptoms iii) Signs d) Secondary Glaucoma: Causes 6. Principles of Management: a. Pharmacological treatment. b. Surgical Management: c. Laser treatment	4 hrs

Learning objectives	Contents covered in this topic	Teaching Hours
Student will be able to: 1. describe the gross anatomy of the retina and its function 2. describe the normal fundus. 3. describe the fundal features of diabetic, hypertensive retinopathy. 4. examine normal eye with use of direct ophthalmoscope 5. identify or suspect vitreous retinal disorder and refer patient	Retina and vitreous: 1. Gross Anatomy: i. Vitreous ii. Retina 2. Function of retina. i. Normal vision. (acuity of vision) ii. Color vision 3. Symptoms Suggestive of vitreous- retinal disorder. 4. Examination of normal eye with direct ophthalmoscope. 5. Fundal features of a. Diabetic retinopathy. b. Hypertensive retinopathy. 6. Referral criteria a. Abnormal red reflex of fundus b. Visual loss or symptoms	3 hrs
Student will be able to: 1. define the common refractive errors eg. myopia, hypermetropia & astigmatism. 2. define Aphakia and pseudophakia 3. define presbyopia and describe the rule of thumb for correction of presbyopia 4. demonstrate basic knowledge about contact lens and refractive surgery. 5. define low vision and mention importance of low vision aid for rehabilitation. 6. record visual acuity. 7. do prescription of presbyopic glass as per rule of thumb and referring difficult patients to ophthalmologists. 8. refer all cases for final correction by ophthalmologist 9. detection of cases with low vision and refer to low vision aid centers	Refraction, Contact lens, Refractive Surgery and Low vision (Gross idea): 1. Refractive status& management a. Emmetropia. b. Myopia. c. Hypermetropia. d. Astigmatism. e. Presbyopia f. Aphakia- I. Spectacle correction II. Contact lens III. Intraocular lens and pseudophakia IV. Refractive surgery (Basic idea) 6. Low vision. Definition of low vision. Refer to low vision aid centre	3 hrs

Learning objectives	Contents	Teaching Hours
Students will be able to. 1. name tumors affecting the eye and adnexa 2. name the causes of leucokoria in children. 3. describe stages, symptoms, signs and management of retinoblastoma 4. diagnose Leucokoria and mention its importance for early referral	Leucocoria in children a. Cataract b. Retinoblastoma c. Endophthalmitis d. Persistent fetal vasculature (PVF/PHPV) e. Retinopathy of prematurity	1 hrs
Student will be able to: 1. describe Strabismus. 2. describe the importance of measuring visual acuity of children of two to five years old 3. describe the causes of amblyopia in children 4. describe the causes of Leukocoria 5. demonstrate the skill of: a. recording visual acuity in children b. ocular motility test c. recognize strabismus, nystagmus and amblyopia for immediate specialist referral.	Ocular motility and paediatric ophthalmology: 1. Gross Anatomy. Extra-ocular muscles 2. Amblyopia.- Definition, cause & impact 3. Strabismus/squint: Definition, cause, diagnosis, effects and management principle 4. Nystagmus: Definition & identification	2 hrs

Learning objectives	Contents	Teaching Hours
Student will be able to: 1. describe visual and pupillary, path ways. 2. describe manifestations of III, IV & VI cranial nerve palsy. 3. describe Papilloedema 4. record visual acuity. 5. perform confrontation visual field testing in four quadrants for each eye. 6. examine pupillary light reflex 7. recognize and diagnose nystagmus. 8. examine the optic disc with the direct ophthalmoscope	Optic Nerve and Neuro Ophthalmology: A. Gross Anatomy 1. Visual path way. 2. Pupillary Pathway B. Examination procedure: 1. VA 2. Visual field testing (confrontation) 3. Pupillary light reflex. 4. Direct Ophthalmoscopy	2 hrs
Student will be able to: 1. describe types of ocular injury 2. explain the effect of different types of ocular trauma 3. mention criteria for referral of the patients 4. demonstrate skill of: a) examination of the eye to assess the effect of injury b) removal of superficial conjunctival, sub-tarsal and superficial corneal foreign body c) performing pad-bandage of the eye d) providing primary management of ocular trauma e) referring the patient after primary management to ophthalmologist /hospital	Ocular trauma: 1. Blunt injury (Details) 2. Perforating Injury. 3. Foreign Body:(Extra and intra ocular) 4. Chemical Injury (details) 5. Thermal injury (Basic idea) 6. Radiation injury (Basic idea)	2 hrs

Learning objectives	Contents	Teaching Hours
Students will be able to: - describe fundal change in hypertension - describe fundal change in diabetes mellitus. - describe ocular manifestation of vitamin-A deficiency and management. - provide health education regarding importance of yearly eye checkup by ophthalmologist for prevention of blindness due to diabetes. - demonstrate the skill of detecting disc oedema on fundus examination with direct ophthalmoscope - recognize Bittot's spot, xerophthalmia and Kerotomalacia & referral.	Ocular Manifestations of systemic diseases (Gross idea): - Diabetes mellitus - Hypertension - Vitamin A Deficiency - Auto-immune diseases (Basic idea) - Tuberculosis - AIDS	2 hrs
Student will be able to: - describe etiology, magnitude and impact of blindness. - demonstrate the concept of 'Primary Eye care' - describe Ocular hygiene. - describe diseases and conditions for referral. - describe concept of school sight test. - define low vision - demonstrate gross idea about communicable and preventable eye diseases. - perform school sight test - identify cases of low vision and referral. - implement "Primary Eye Care" concept at the place of work - develop awareness about eye donation in the community. - diagnose & initiate initial management of ocular emergency	Miscellaneous & Community eye care: - Etiology and magnitude of blindness - School sight test. - Primary eye care - Referral guide line - Low vision and rehabilitation - Outreach activities. - Eye donation & eye banking. - Vision 2020, The right to sight (Gross idea) - Ocular therapeutics - Ocular emergency - Sudden loss of vision - Painful loss of vision - Painless loss of vision - Gradual dimness of vision - Red eye - Ocular effects of environmental change	5 hrs

EXAMINATION SKILLS	Skills-		Assist	Observe
	Able to perform Independently	Able to Perform under Guidance		
1. Visual Acuity test and Use of pinhole (including light perception, projection)	✓			
2. Colour Vision test		✓		
3. Visual field by confrontation	✓			
4. Examination of ocular movements	✓			
5. Fluorescein staining to identify corneal abrasion		✓		
6. Pupillary size and reaction	✓			
7. Distant direct ophthalmoscopy on dilated pupils to diagnose lens opacities		✓		
8. Method of Direct ophthalmoscopy		✓		
9. Digital tonometry	✓			
10. Schiottz tonometry				✓
11. Regurgitation for NLD Block	✓			
12. Syringing				✓
13. Instillation of eye drops/ ointment	✓			
14. Irrigation of conjunctiva	✓			
15. Applying of patching	✓			
16. Epilation of cilia		✓		
17. Eversion of upper eye lid	✓			
18. Removal of corneal foreign body				✓
19. Cataract surgery				✓
20. Glaucoma surgery				✓
21. Chalazion/Stye				✓
22. Tarsorrhaphy			✓	
23. Assessment of Opacity in the media	✓			
24. Lacrimal Sac Surgery				✓